

Yasin Torres-Tiji
Colorado School of Mines – Chemistry Department
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Professional Summary:

Algal synthetic and systems biologist with end-to-end expertise spanning genomic analysis, gene design, transformation, bioreactor operation, and multi-omics characterization. Currently engineering *Picochlorum celeri* at the systems level as a postdoctoral scholar at Colorado School of Mines, with active collaborations across DOE national laboratories (PNNL, LANL, NLR, EMSL). PI on a DOE/EMSL campaign integrating transcriptomics, proteomics, and metabolomics to build predictive genome-scale metabolic models of algal productivity. Seeking faculty, PI, or staff scientist positions.

Education

University of California San Diego

Division of Biological Science
Ph.D. Molecular Biology - 2021

Autonomous University of Barcelona

B.S. Biotechnology – 2015

Additional Education

University of California San Diego

MicroMBA Program — 2023

Business analytics, corporate finance, marketing, operational strategy, and leadership.

National Center for Genome Resources

Differential Gene Expression Workshop — 2025

RNA-seq experimental design, read alignment, statistical analysis with DESeq2, and pathway enrichment.

Codecademy

Learn Python 3 — 2026

Programming fundamentals, data structures, control flow, functions, and file handling in Python.

Research Interests

Algal synthetic & systems biology · Genetic engineering · Multi-omics integration & genome-scale metabolic modeling · Bioreactor process engineering & climate simulation · Photosynthetic carbon metabolism & partitioning · Sustainable bioproduction & bioeconomy · AI/ML-assisted bioinformatics

Dissertation: “Optimizing recombinant protein production in *Chlamydomonas reinhardtii*: turning a model organism into a commercially relevant biotechnological host

Performed research in recombinant protein production in microalgae with expertise throughout the whole process: gene design, gene cloning, microalgal transformation with heterologous DNA, microalgal culturing, recombinant protein detection, recombinant protein purification and analysis. Focused on the optimization of two crucial elements of the process: improvement of recombinant protein production at single cell level by developing advanced genetic tools, and improvement of recombinant protein production by increasing the cell density in a heterotrophic bioreactor.

Skills

Molecular & Synthetic Biology

- End-to-end recombinant protein production: codon-optimized gene design, expression vector assembly, algal/bacterial transformation, high-expression clone selection, bioreactor scale-up, and protein purification.
- Gene design, molecular cloning, PCR, RT-qPCR, and nucleic acid extraction/purification.
- Genome editing (CRISPR), transgenesis, and cis-regulatory element identification in photosynthetic organisms.
- Western blot, chromatography, FPLC, ELISA, and mass spectrometry-based protein characterization.

Multi-Omics & Bioinformatics

- RNA-seq: experimental design, library preparation, STAR alignment, featureCounts, DESeq2 differential expression analysis, GO/KEGG pathway enrichment.
- TMT quantitative proteomics: LC-MS/MS data processing via FragPipe/MSFragger, limma-based differential expression analysis, TMT-Integrator for ratio quantification.
- Ubiquitination proteomics: site-level quantification and differential analysis (5,660+ sites), protein-abundance normalization, condition-specific detection workflows.
- Metabolomics: LC-MS (HILIC) and GC-MS data interpretation and integration.
- Genome annotation: multi-source pipelines integrating CLEAN, InterProScan, SwissProt BLAST, and eggNOG-mapper for functional characterization of non-model organisms.
- Genome-scale metabolic model construction and constraint-based analysis.
- Proficient in R and Python for bioinformatic analysis, statistical computing, and data visualization. Working environment: WSL2 Ubuntu with conda-managed pipelines (STAR, fastp, BLAST+, Diamond, samtools, bedtools).
- NGS data analysis, genome assembly, and variant calling using command-line tools and Galaxy.
- Flow cytometry experimental design and data analysis (FlowJo).

Analytical Chemistry

- FAME extraction and fatty acid profiling by GC-FID.
- ELISA-based protein quantification.
- Mass spectrometry data analysis and interpretation (LC-MS/MS, GC-MS).

Microscopy & Microbial Culturing

- Optical, fluorescence, and confocal microscopy.
- Microalgal cultivation across scales: agar plates, flasks, open pond systems, and photobioreactors.
- Microbial identification via 16S/ITS/18S sequencing, metagenomics, and whole-genome sequencing.
- Sterile technique and contamination-free culture maintenance.

Bioreactor Process Engineering

- Design, operation, and media optimization of batch and semi-continuous bioreactor systems.
- Mastery in operating climate-simulated photobioreactors for generating reproducible growth data under controlled light, temperature, and CO₂ regimes.
- Process optimization for biomass titer and recombinant protein yield.

Communication, Mentorship & Project Management

- PI on DOE/EMSL CAM campaign; experienced in grant writing, project planning, and execution.
- Mentored 15+ undergraduate students and 4 graduate students.
- Effective communicator of complex scientific concepts to technical and non-technical audiences.
- Peer reviewer for 9 journals including Microbial Cell Factories, Food Bioscience, PLOS ONE, and Frontiers in Plant Science.

Professional Experience

Postdoctoral Fellow - Posewitz Lab

Colorado School of Mines, Department of Chemistry, Golden, CO

July 2024 – Present

- Performing research in algal synthetic biology and algal bioprocess engineering to enhance the cost competitiveness of algal bioproducts.

Postdoctoral Scholar - Mayfield Lab

University of California San Diego, Division of Biological Sciences, La Jolla, CA

October 2021 – June 2024

- Performed research in synthetic biology to create genetic tools that perfect our capacity to genetically engineer microalgae for the improved production of biofuels, bioplastics precursors, food and therapeutics.

Technical achievements: Successfully identified a series of DNA cis-regulatory elements using bioinformatics analysis, significantly enhancing transgene expression. When integrated into a minimal promoter, these elements not only matched but surpassed the efficacy of current state-of-the-art genetic tools in synthetic promoter design.

PhD Student - Mayfield Lab

University of California San Diego, Division of Biological Sciences, La Jolla, CA
September 2015 – September 2021

- Performed research developing genetic tools to improve transgene expression in the nuclear genome of *Chlamydomonas reinhardtii* and advanced the field of recombinant protein production in bioreactors using algae.
Technical achievements: 2-fold improvement on biomass yield in a bioreactor, 3-fold improvement on recombinant protein titers compared to highest reported in the literature. Novel genetic tool producing a 10-fold improvement in transgene expression compared to state of the art.

Research Assistant - Mayfield Lab

University of California San Diego, Division of Biological Sciences, La Jolla, CA
January 2015 – June 2015

- Performed research developing genetic tools for recombinant protein production in the algal chloroplast of *Chlamydomonas reinhardtii*.

Research Assistant

Center for Research in Agricultural Genomics (CRAG), Barcelona, Spain
August 2012 – May 2014

- Performed research revolving the genetic regulation of lignin formation in corn to make second generation biofuels more economically viable.

Funding & Awards

U.S. Department of Energy (DOE) - Environmental Molecular Sciences Laboratory (EMSL)

Grant No. 62132 (Feb 2026–Present)

Project: Genome-Scale Modeling of *Picochlorum celeri* to Elucidate CO₂-Dependent Extreme Growth and Carbon Partitioning in Algae

Richland, WA, USA. DOI: 10.46936/camc.proj.2026.62132/60015765

Role: Principal Investigator

Grant Banco Santander Formula 2014/2015 (5000 EUR award for studying abroad - 1 out of 3 recipients for the Autonomous University of Barcelona)

University of California Education Abroad Program 2014/2015 (1 out of 25 recipients for the Autonomous University of Barcelona)

Frontiers of Innovation Scholars Program 2016/2017 (25,000 USD fellowship University of California San Diego)

Publications

Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2020). Microalgae as a future food source. *Biotechnology Advances*, 41, 107536. <https://doi.org/10.1016/j.biotechadv.2020.107536>
Citations: 626

Diaz, C. J., Douglas, K. J., Kang, K., Kolarik, A. L., Malinovski, R., Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2023). Developing algae as a sustainable food source. *Frontiers in Nutrition*, 9, 3147. <https://doi.org/10.3389/fnut.2022.1029841> **Citations: 189**

Berndt, A. J., Smalley, T. N., Ren, B., Simkovsky, R., Badary, A., Sproles, A. E., Torres-Tiji, Y., Melis, A., & Mayfield, S. P. (2021). Recombinant production of a functional SARS-CoV-2 spike receptor binding domain in the green algae *Chlamydomonas reinhardtii*. *PloS One*, 16(11), e0257089. <https://doi.org/10.1371/journal.pone.0257089> **Citations: 53**

Torres-Tiji, Y., Fields, F. J., Yang, Y., Heredia, V., Horn, S. J., Keremane, S. R., Hannon, M. J., & Mayfield, S. P. (2022). Optimized production of a bioactive human recombinant protein from the microalgae *Chlamydomonas reinhardtii* grown at high density in a fed-batch bioreactor. *Algal Research*, 66, 102786. <https://doi.org/10.1016/j.algal.2022.102786> **Citations: 45**

Gupta, A., Kang, K., Pathania, R., Saxton, L., Saucedo, B., Malik, A., Torres-Tiji, Y., Diaz, C. J., Dutra Molino, J. V., & Mayfield, S. P. (2024). Harnessing genetic engineering to drive economic bioproduct production in algae. *Frontiers in Bioengineering and Biotechnology*, 12, 1350722. <https://doi.org/10.3389/fbioe.2024.1350722> **Citations: 42**

Torres-Tiji, Y., Sethuram, H., Gupta, A., McCauley, J., Dutra-Molino, J.V., Pathania, R., Saxton, L., Kang, K., Hillson, N. J., & Mayfield, S. P. (2024) Bioinformatic prediction and high throughput in vivo screening to identify cis-regulatory elements for the development of algal synthetic promoters. *ACS Synthetic Biology*, 13(7), 2150–2165. <https://doi.org/10.1021/acssynbio.4c00199> **Citations: 10**

Molino, J. V. D., Oliver, A., Sethuram, H., Kang, K., Saucedo, B., Diaz, C. J., Gupta, A., Jen, L. J., Torres-Tiji, Y., Hidasi, N., Badary, A., Jenkins, H., Fields, F. J., Simkovsky, R., & Mayfield, S. P. (2025) Description of a novel extremophile green algae, *Chlamydomonas pacifica*, and its potential as a biotechnology host. *Algal Research*, 88, 104034. <https://doi.org/10.1016/j.algal.2025.104034> **Citations: 7**

Torres-Tiji, Y., & Posewitz, M. C. (2025) Biotechnology of photosynthetic algae for a world demanding sustainable fuels and resources. *One Earth*, 8(7), 101381. <https://doi.org/10.1016/j.oneear.2025.101381> **Citations: 2**

Gupta, A., Oliver, A., Dutra Molino, J. V., Wnuk-Fink, K. M. J., Tessman, M., Kang, K., do Espirito Santo, É., Torres-Tiji, Y., Burkart, M. D., & Mayfield, S. P. (2025) Identification and overexpression of endogenous transcription factors to enhance lipid accumulation in the biotechnologically relevant species *Chlamydomonas pacifica*. *Algal Research*, 90, 104192. <https://doi.org/10.1016/j.algal.2025.104192> **Citations: 0**

Torres-Tiji, Y., Fields, F. J., & Mayfield, S. P. (2026). Microalgal colony blot: A simple and rapid method for direct detection of recombinant protein production in microalgae colonies. *PLOS ONE*, 21(3), e0345458. <https://doi.org/10.1371/journal.pone.0345458> **Citations: 0**

Eknesh, T. S., Karns, D. A. J., Torres-Tiji, Y., MacLaughlin, D., Crowe, B., Martinez, J., Benemann, J. R., Posewitz, M. C., & Weissman, J. C. (2026) Biomass Productivity of the

Marine Alga *Picochlorum celeri* at Different Carbon Dioxide and Oxygen Concentrations with pH Cycling. *Journal of Environmental Chemical Engineering* (**In Submission**)

Doctoral Thesis

Torres-Tiji, Y. (2021). Optimizing Recombinant Protein Production in *Chlamydomonas reinhardtii*: Turning a Model Organism into a Commercially Relevant Biotechnological Host. Doctoral Thesis, University of California, San Diego.

Other works

Torres Tiji, Y. (2015). Viability analysis of algae biofuels and assessment of the future energy situation.

Conferences/Public Talks

Synthetic Biology: Engineering, Evolution & Design – 2026 - Poster

Symposium on Biomaterials, Fuels and Chemicals – 2026 – Selected Talk

EMSL-JGI Joint User Meeting – 2026 - Poster

U.S. Department of Energy – Algae Productivity Exceeding Expectations Challenge Meeting - 2023

Algal Biomass, Biofuels and Bioproducts – 2023 - Poster

6th International Congress Meredith Gould – 2023 - Presentation

PepTalk – The Protein Science and Production Week – 2022 – Presentation

Algae Biomass Organization Summit 2020 – Presentation

Frontiers of Innovations Scholars Program – 2017 - Presentation

Peer Review Contributions

ACS Syn Bio – 1 review (2026)

Algal Research — 1 review (2026)

Journal of Functional Foods — 1 review (2026)

Microbial Cell Factories (Springer Nature) — 1 review (2026)

Food Bioscience (Elsevier) — 1 review (2025)

Journal of Agricultural and Food Chemistry (American Chemical Society) — 1 review (2025)

Journal of Biotechnology (Elsevier) — 1 review (2025)

Frontiers in Genome Editing (Genome Editing in Plants) — 1 review (2025)

Frontiers in Plant Science (Plant Biotechnology) — 1 review (2025)

Limnology and Oceanography: Methods (ASLO) — 1 review (2024)

PLOS ONE — 2 reviews (2024)

Critical Reviews in Biotechnology — 1 review (2020)

Mentorships

ENLACE Mentorship Program - Summer 2019

ENLACE Mentorship Program – Summer 2023

Mentored over 15 undergraduate students, and 4 graduate students.